

REGULATION OF SECRETION OF TH

Secretion of thyroid hormones controlled by ant. pituitary and hypothalamus through feedback mechanism.

1. ROLE OF PITUITARY GLAND

Thyroid-Stimulating Hormone (TSH)

- TSH secreted by ant. pituitary.
- Also necessary for growth and secretory activity of thyroid gland.
- TSH influences every stage of formation & release of thyroid hormone.
- Chemistry → Peptide hormone with 1 α -chain & 1 β -chain.
- Plasma Level → Normal plasma level is approximately $2 \mu\text{U/mL}$.
- Half-life → About 60 minutes.

Actions of TSH

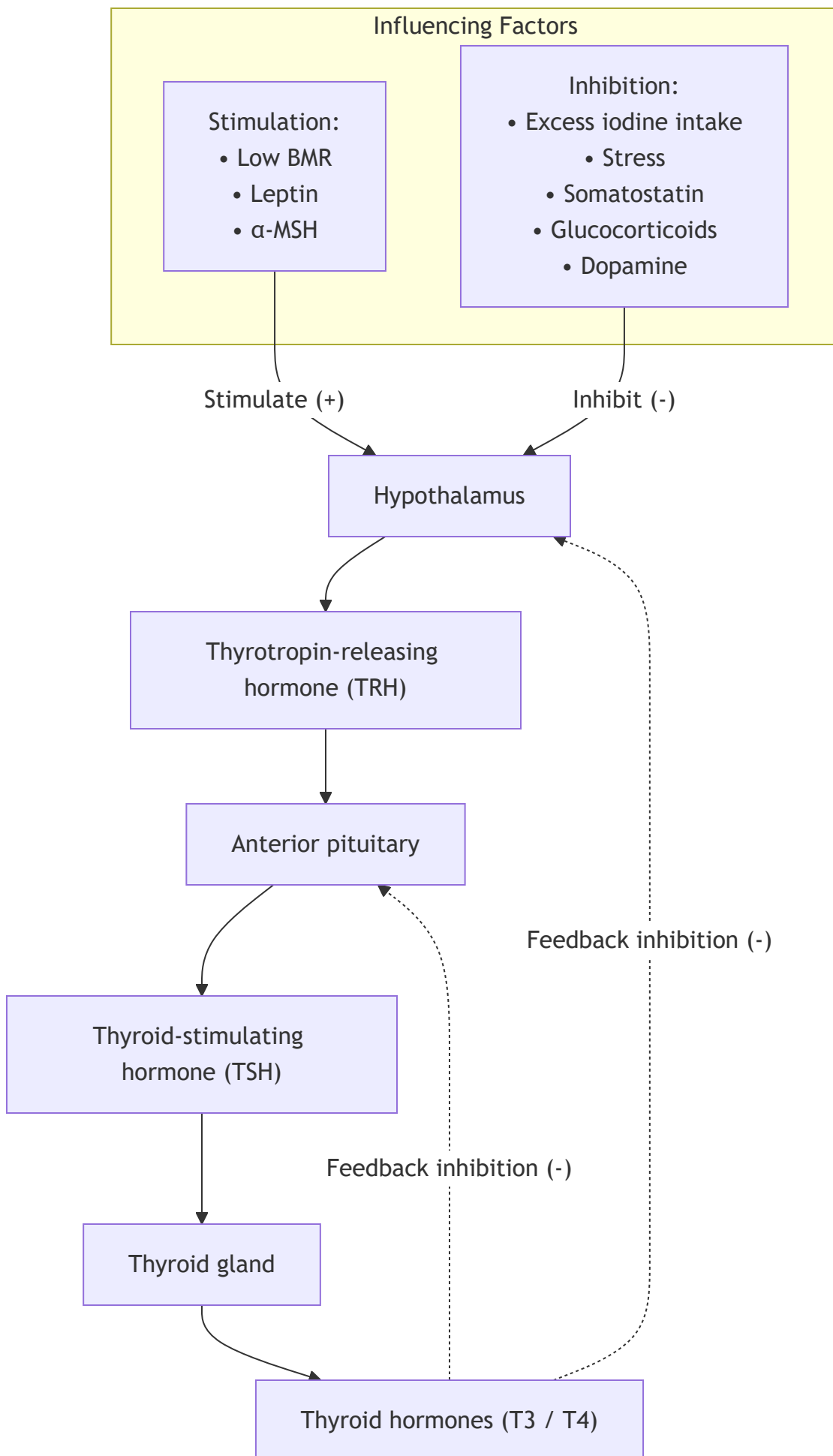
- TSH ↑ses:
 - a. No. of follicular cells of thyroid.
 - b. Conversion of cuboidal cells into columnar cells in thyroid gland:
 - ↓
 - Causes development of thyroid follicles.
 - c. Size and secretory activity of follicular cells.
 - d. Iodide pump & iodide trapping in follicle cells.
 - e. Thyroglobulin secretion into follicles.
 - f. Iodination of tyrosine & coupling to form hormone.
 - g. Proteolysis of thyroglobulin.
- Mode of Action of TSH → Acts through cyclic AMP mechanism.

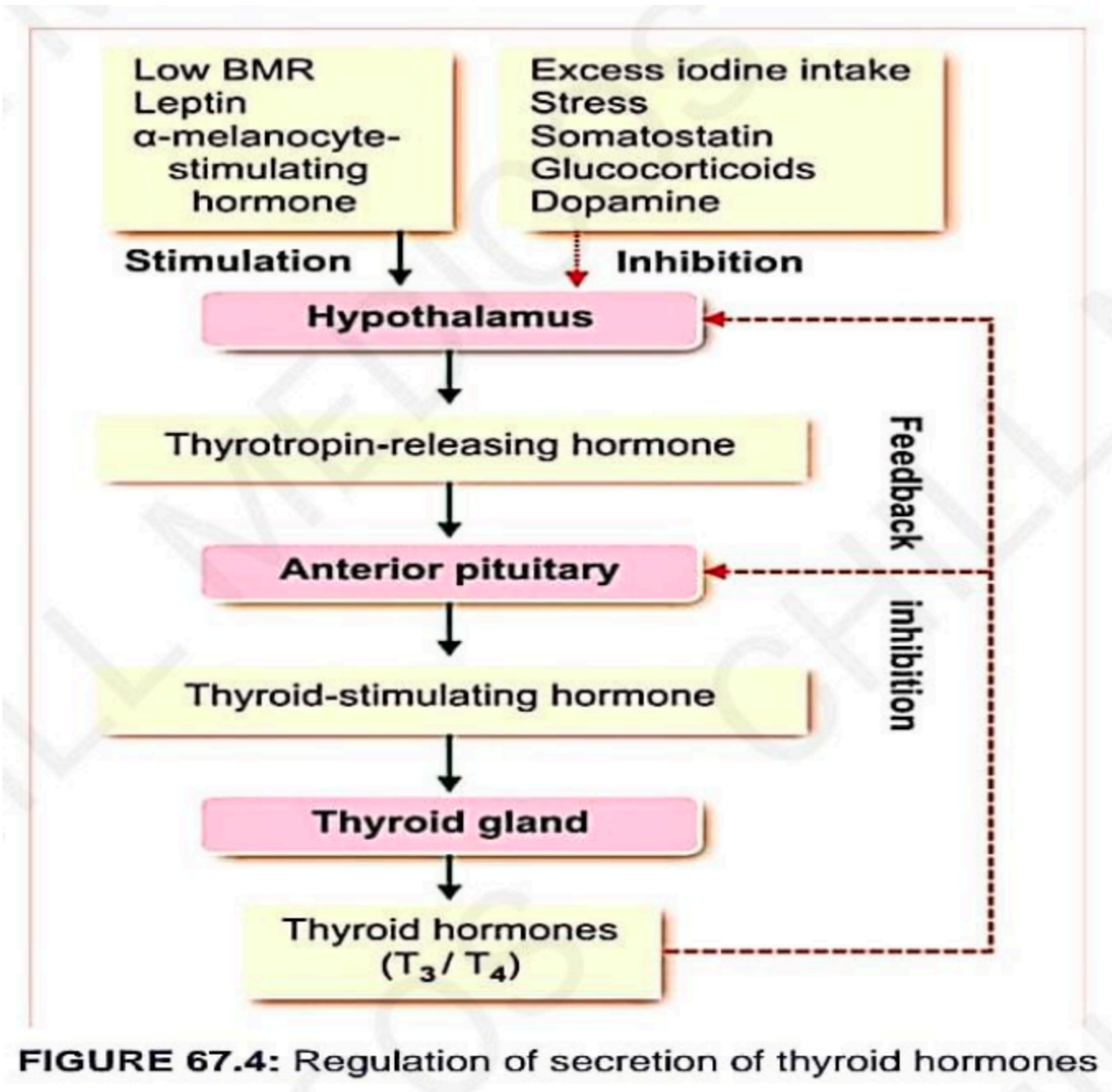
2. ROLE OF HYPOTHALAMUS

- **Hypothalamus regulates thyroid secretion** through **Thyrotropic-Releasing Hormone (TRH)**.
- **TRH transported through** hypothalamo-hypophyseal portal vessel:
 - ↓
 - To ant. pituitary
 - ↓
 - Cause release of TSH.

3. FEEDBACK CONTROL

- **Thyroid hormones regulate their own secretion** through **-ve feedback control**:
 - ↓
 - Inhibiting release of TRH from hypothalamus and TSH from anterior pituitary.





4. ROLE OF IODIDE

- When iodine intake is high, enzyme necessary for synthesis of thyroid hormones **inhibited by iodide itself**:
- ↓
- Result in **suppression of hormone synthesis**
- ↓
- Effect of iodide called **Wolff-Chaikoff effect**.

5. ROLE OF OTHER FACTORS

Factors ↑sing Secretion of TH:

- a. Low BMR
- b. Leptin
- c. α -Melanocyte stimulating hormone (α -MSH)

Leptin (*from adipose tissue*) and **α -melanocyte stimulating hormone** (*from pituitary*):

↓

↑se release of TRH & synthesis of T_4 .

Factors ↓sing Secretion of TH:

(*by inhibiting release of TRH*)

- a. Excess iodide intake
- b. Stress
- c. Somatostatin
- d. Glucocorticoids
- e. Dopamine